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QUESTIONS

Q1 ✓
Define a graph.
2 marks**Q2** ✓
What are the major types of graphs?
2 marks**Q3** ✓
Differentiate between directed and undirected graphs.
2 marks**Q4** ✓
What is a weighted graph?
2 marks**Q5** ✓
What is a topological sort?
2 marks**Q6** ✓
A directed graph has the dependencies $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.
2 marks**Q7** ✓
An unweighted graph has vertices A, B, C, D, E, F with edges A-B, A-C, B-D, C-D, C-E.**Q1 Define a graph.**

2 marks

Define a graph.

Your Answer

A graph is data structure consisting of nodes and edges that connects a pair of vertices

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An unweighted graph has vertices A, B, C, D, E, F with edges A-B, A-C, B-D, C-D, C-E.
2 marks

Q2 What are the major types of graphs?

2 marks

What are the major types of graphs?

Your Answer

Directed graph
Undirected graph

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2 marks

Q3 Differentiate between directed and undirected graphs.

2 marks

Differentiate between directed and undirected graphs.

Your Answer

Directed graph: Edge have a specific direction
undirected graph: Edge have no direction

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An unweighted graph has vertices A, B, C, D, E, F with edges A-B, A-C, B-D, C-D, C-E.**Q4** What is a weighted graph?

2 marks

What is a weighted graph?

Your Answer

A weighted graph is a graph in which each edge has a numerical values called weight

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An unweighted graph has vertices A, B, C, D, E, F with edges A-B, A-C, B-D, C-D, C-E.
2 marks**Q5 What is a topological sort?**

2 marks

What is a topological sort?

Your Answer

A topological sort is a linear ordering of vertices.

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Q4

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Q5

What is a topological sort?
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Q6

A directed graph has the dependencies $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.
2 marks



Q7

An unweighted graph has vertices A, B, C, D, E, F with edges A-B, A-C, B-D, C-D, C-E.
2 marks



Q6 A directed graph has the dependencies $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid. 2 marks

A directed graph has the dependencies $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $B \rightarrow E$, and $E \rightarrow F$. Determine a valid topological ordering and explain why the ordering is valid.

Your Answer

Valid topological ordering: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F$
 A comes before B: $A \rightarrow B$
 B comes before C: $B \rightarrow C$
 C comes before D: $C \rightarrow D$
 D comes before E: $D \rightarrow E$
 E comes before F: $E \rightarrow F$

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Q7 
An unweighted graph has vertices A, B, C, D, E, F with edges A-B, A-C, B-D, C-D, C-E, D-F, and E-F. Starting from A, determine the shortest path to F using BFS.
2 marks

Q7 An unweighted graph has vertices A, B, C, D, E, F with edges A-B, A-C, B-D, C-D, C-E, D-F, and E-F. Starting from A, determine the shortest path to F using BFS. 2 marks

An unweighted graph has vertices A, B, C, D, E, F with edges A-B, A-C, B-D, C-D, C-E, D-F, and E-F. Starting from A, determine the shortest path to F using BFS.

Your Answer

Shortest Path: A->C->D->F

 Save Answer